

An endpoint-majority condition for three-point Grünbaum rate–distortion maximizers

Xiaojian Zeng

PBC School of Finance, Tsinghua University, 100190

xiaojianzeng@foxmail.com

Abstract

For a finitely supported real law P , let $D(P)$ be the largest squared transport cost between two copies of P , and put $\alpha(P) = R_P(D(P)/4)$, with information measured in nats. We prove that every genuine three-point law whose two endpoint masses are at most $1/2$ satisfies $\alpha(P) < \log(23209/20000) < 0.14882 < \alpha_3$, where α_3 is the supremum over laws on two or three points. Consequently every maximizing law has an endpoint atom of mass greater than $1/2$. The proof gives a uniform bound for a three-Gaussian envelope, combining analytic tails with an exact rational certificate covering 33,335 parameter rectangles. We include the duality, a separating rational witness, and the boundary argument ensuring attainment. The exact value of α_3 and the full classification of its maximizers remain unresolved by this note; the envelope bound is not asserted to be sharp.

1 Statement and scope

For a nonconstant finitely supported real random variable with law P , define

$$D(P) = \sup_{X_1 \sim P, X_2 \sim P} \mathbb{E}(X_1 - X_2)^2, \quad R_P(\Delta) = \inf_{\mathbb{E}(X-Y)^2 \leq \Delta} I(X; Y), \quad (1)$$

where the first supremum is over all couplings with the indicated marginals, and the reconstruction Y is real valued. Write

$$\alpha(P) = R_P(D(P)/4), \quad \alpha_2 = \sup_{|\text{supp } P|=2} \alpha(P), \quad \alpha_3 = \sup_{2 \leq |\text{supp } P| \leq 3} \alpha(P).$$

All logarithms are natural. Translation and nonzero scaling leave α unchanged, since they preserve information and scale both squared costs by the same factor. A genuine three-point law can therefore be written as

$$P = a\delta_0 + b\delta_x + c\delta_1, \quad 0 < x < 1, \quad a, b, c > 0, \quad a + b + c = 1. \quad (2)$$

Theorem 1. *Every law (2) with $a, c \leq 1/2$ satisfies*

$$\alpha(P) < \log \frac{23209}{20000} < 0.14882 < \alpha_3. \quad (3)$$

The supremum α_3 is attained by a genuine three-point law. Every law attaining it has an endpoint atom of mass strictly greater than $1/2$.

The motivating rate–distortion formulation is attributed to Arman, Bondarenko, Prymak and Radchenko [1]. The supplied problem description reports a solved binary optimization and a numerical three-point improvement. The contribution established here is a uniform exclusion of both central transportation chambers, giving a necessary condition on every maximizer. It neither evaluates the remaining maximum nor identifies all maximizers. No priority claim is made. As adjacent computational methodology, Zeng et al. [2] also use exact certificates on compact parameter regions; that reference is not used as a rate–distortion result.

The analytic statement underlying the exclusion is the following.

Theorem 2. *Let $0 \leq u \leq v$ and $\nu_0, \nu_1, \nu_2 \geq 0$. If*

$$G(t) = \nu_0 e^{-t^2} + \nu_1 e^{-(t-u)^2} + \nu_2 e^{-(t-v)^2} \leq 1 \quad (t \in \mathbb{R}), \quad (4)$$

then

$$B := \nu_0 e^{-u^2/2} + \nu_1 + \nu_2 e^{-(v^2-u^2)/2} < K := \frac{23209}{20000}. \quad (5)$$

The proof produces one common bound strictly below K for the whole unbounded domain. This uniformity is needed when taking the supremum in the rate–distortion dual; pointwise strictness alone would not suffice.

2 Transport and Gaussian duality

Maximizing the squared transport cost is equivalent to minimizing $\mathbb{E}X_1X_2$. If a coupling places positive mass on two increasing pairs, swapping a common amount ε to the crossed pairs decreases the product cost by $\varepsilon(z' - z)(y' - y) > 0$. Successively matching the lowest remaining first-coordinate mass with the highest remaining second-coordinate mass therefore gives the optimal antitone quantile coupling. For (2), its off-diagonal entries are

$$\pi_{01} = \min(a, c), \quad \pi_{0x} = (\min(a, 1 - a) - c)_+, \quad \pi_{x1} = (\min(c, 1 - c) - a)_+.$$

These entries occur symmetrically. Thus the four closed chambers, agreeing on their common boundaries, give

$$\frac{D(P)}{2} = \begin{cases} c + bx^2, & a \geq 1/2, \\ a + b(1 - x)^2, & c \geq 1/2, \\ ax^2 + c(1 - x^2), & a, c \leq 1/2, \ a \geq c, \\ a(2x - x^2) + c(1 - x)^2, & a, c \leq 1/2, \ c \geq a. \end{cases} \quad (6)$$

The last two cases are the central chambers of Theorem 1.

Lemma 3. *For a finite law $P = \sum_i p_i \delta_{z_i}$ with $p_i > 0$ and $\Delta > 0$,*

$$R_P(\Delta) = \sup_{s, \nu} \left\{ H(p) + \sum_i p_i \log \nu_i - s\Delta \right\}, \quad (7)$$

where $s \geq 0$, $\nu_i > 0$, $H(p) = -\sum_i p_i \log p_i$, and $\sum_i \nu_i e^{-s(z_i - y)^2} \leq 1$ for every $y \in \mathbb{R}$.

Proof. Clip reconstructions to the input interval J ; distortion cannot increase and information cannot increase. For fixed finite $s \geq 0$, the penalized value $L_s = \inf(I + s\mathbb{E}(X - Y)^2)$ satisfies

$$L_s = \min_Q \left[-\sum_i p_i \log m_i(Q) \right], \quad m_i(Q) = \int_J e^{-s(z_i - y)^2} dQ(y). \quad (8)$$

Indeed $I = \inf_Q \text{KL}(P_{XY} \| P_X Q)$; interchanging these two infima and minimizing each conditional row gives its normalized Gibbs density. The possible vectors $m(Q)$ form the compact convex hull of the continuous Gaussian curve on J . Their coordinates are bounded away from zero for fixed s , so a minimum m^* exists. The directional derivative of the convex objective toward any curve point gives, with $\nu_i = p_i/m_i^*$,

$$C(y) := \sum_i \nu_i e^{-s(z_i - y)^2} \leq 1 \quad (y \in J).$$

It also holds outside J : for $s > 0$ the sum increases to the left of J and decreases to its right, while for $s = 0$ it is constant. A representing measure Q for m^* is supported on $C = 1$, since $\int C dQ = 1$. Hence the Gibbs channel has output marginal Q , and its value is $H(p) + \sum_i p_i \log \nu_i$. A finite representing measure exists by the finite-dimensional convex-hull theorem; four points suffice for three inputs.

Conversely, for any feasible positive ν and any posterior vector r at reconstruction y , apply relative-entropy nonnegativity to $q_i = \nu_i e^{-s(z_i - y)^2} / C(y)$. It yields

$$\text{KL}(r \| p) + s \sum_i r_i (z_i - y)^2 \geq \sum_i r_i \log(\nu_i / p_i) - \log C(y).$$

Integrating proves the matching lower bound for L_s and, at distortion at most Δ , the lower bound in (7).

Finally R_P is finite, convex and nonincreasing on nonnegative distortion, and continuous at every positive distortion. Convexity follows by using an input-independent time-sharing flag and then discarding it. Supporting lines at $\Delta > 0$ give $R_P(\Delta) = \sup_{s \geq 0} (L_s - s\Delta)$. At or above the variance, constant reconstruction gives $R_P = 0$ and $s = 0$ suffices. This proves the asserted equality, rather than only a dual lower bound. $\square$

3 The binary comparison and unbounded tails

Let $h(p) = -p \log p - (1 - p) \log(1 - p)$. For a Bernoulli law, reflection permits $0 < p \leq 1/2$, and the antitone coupling gives $D = 2p$. For $d \in [0, 1/4]$ put

$$g(d) = h\left(\frac{1 - \sqrt{1 - 4d}}{2}\right).$$

This function is increasing and concave: writing $t = \sqrt{1 - 4d}$ gives $g'(d) = 2 \text{atanh}(t)/t$, which decreases with d , by the positive series for $\text{atanh}(t)/t$. If $r = \Pr(X = 1 \mid Y)$, then $h(r) = g(r(1 - r))$, while $\mathbb{E}r(1 - r) \leq \mathbb{E}(X - Y)^2$. Jensen's inequality therefore gives $I \geq h(p) - g(\Delta)$. At $\Delta = p/2$, a posterior taking the two values $q = (1 - \sqrt{1 - 2p})/2$ and $1 - q$, with mean p , attains this bound; such a posterior exists since $q \leq p \leq 1 - q$. Reconstruction by the posterior has distortion $q(1 - q) = p/2$. Consequently

$$f(p) := \alpha(\text{Bern}(p)) = h(p) - h\left(\frac{1 - \sqrt{1 - 2p}}{2}\right). \quad (9)$$

For $0 < p < 1/2$, with $t = \sqrt{1 - 2p}$, differentiation gives

$$\begin{aligned} f'(p) &= \log \frac{1 - p}{p} - \frac{1}{2t} \log \frac{1 + t}{1 - t}, \\ f''(p) &= \frac{3p - 1}{2p(1 - p)t^2} - \frac{\text{atanh}(t)}{t^3} \leq -\frac{p + 1}{2p(1 - p)} < 0, \end{aligned}$$

where $\operatorname{atanh}(t) \geq t$ proves the inequality. Both endpoint limits of f are zero. A single global tangent, at the rational parameters

$$t_0 = \frac{757817}{1000000}, \quad p_0 = \frac{1 - t_0^2}{2} = \frac{425713394511}{2000000000000},$$

therefore gives

$$\alpha_2 \leq f(p_0) + \max(p_0, 1/2 - p_0)|f'(p_0)| < 0.148624 < \log K_2, \quad K_2 := \frac{1160237}{1000000}. \quad (10)$$

The strict scalar comparisons are evaluated by the rational logarithm series described in Section 6. The tangent bound is less than 0.148623265596.

If a feasible two-Gaussian envelope has centers $0, d$, then

$$\eta_0 + \eta_1 e^{-d^2/2} \leq e^{\alpha_2} < K_2. \quad (11)$$

For positive weights and $d > 0$, choose binary masses proportional to the two summands on the left. Its actual distortion is $D/4 = \min(p_0, p_1)d^2/2 \leq p_1 d^2/2$. The lower bound of Lemma 3, with $s = 1$, is then at least the logarithm of their sum. The definition of α_2 proves (11). A zero weight or $d = 0$ gives a left side at most 1 directly. Reflection gives the same bound with the other endpoint undamped.

For (4), evaluation at each center gives $0 \leq \nu_i \leq 1$. Deleting one Gaussian preserves feasibility. Writing $w = v - u \geq 0$, and using $(v^2 - u^2)/2 = w^2/2 + uw \geq w^2/2$, equation (11) gives

$$B \leq K_2 + e^{-u^2/2}, \quad B \leq K_2 + e^{-(v^2 - u^2)/2}. \quad (12)$$

For $u \geq 5$ the first is at most $K_2 + e^{-25/2}$; for $u \leq 5$ and $v \geq 7$ the second is at most $K_2 + e^{-12}$. Both are strictly less than

$$M := \frac{46417999623}{40000000000} = 1.160449990575 < K, \quad K - M = \frac{377}{40000000000}. \quad (13)$$

Thus it remains to bound B on a compact region. The certificate covers the larger rectangle $(u, w) \in [0, 5] \times [0, 7]$.

4 A finite certificate for the Gaussian bound

For a closed rectangle $Q = [u_l, u_h] \times [w_l, w_h]$, the Gaussian centers are $z_0 = 0$, $z_1 = u$, $z_2 = u + w$. Choose rational pairs (r_j, s_j) and evaluate the universal envelope constraint at $t_j = r_j u + s_j w$. Define

$$A_{ji} = \exp\left(-\max_Q(t_j - z_i)^2\right), \quad (c_0, c_1, c_2) = (e^{-u_l^2/2}, 1, e^{-w_l^2/2 - u_l w_l}). \quad (14)$$

The maximum of the square of an affine function on a rectangle is its maximum over the vertices. Thus the exponents are exact rational numbers. For all shapes in Q , feasibility implies $\sum_i A_{ji} \nu_i \leq 1$, and $B \leq \sum_i c_i \nu_i$.

Lemma 4. *For any nonnegative rational multipliers λ_j , every feasible shape and weight vector over Q satisfies*

$$B \leq U(Q) := \sum_j \lambda_j + \sum_{i=0}^2 \max\left(0, c_i - \sum_j \lambda_j A_{ji}\right). \quad (15)$$

The inequality remains valid if each c_i is replaced by an upper enclosure and each A_{ji} by a lower enclosure.

Proof. Decompose $\sum_i c_i \nu_i$ into the weighted constraints plus their residual coefficients. Each constraint is at most 1 and each weight lies in $[0, 1]$, so a residual contributes at most its positive part. Increasing c_i or decreasing A_{ji} can only increase this upper bound. $\square$

The supplementary file `branch-b-proof-data.json` records 33,335 rational rectangles. On 33,322 of them it supplies constraint indices and nonnegative multipliers with denominator 10^{12} for (15). On the other 13, one of (12), evaluated at lower rectangle endpoints, suffices. The exact verification gives

Quantity	Certified value
Total rectangles	33 335
Multiplier / tail rectangles	33 322 / 13
Area of the domain	35
Distinct u edges / w endpoints	755 / 615
Interior covering multiplicity	1
Common upper enclosure	46417999623/400000000000
Margin below K	377/400000000000

Each rectangle is nondegenerate and contained in $[0, 5] \times [0, 7]$. Coverage is checked by an exact sweep over all vertical edges, adding or removing the corresponding w intervals. On every elementary interval between consecutive w endpoints, both the minimum and maximum cover counts must be 1 between successive u edges, and 0 after $u = 5$. The area check alone is not used as a coverage proof. Since the cover is a finite union of closed rectangles, coverage of the open cells also covers every edge and corner, including the collision boundaries.

All upper bounds are checked with integers and fractions. In particular, there is no inference from the accuracy of the numerical optimizer that proposed the multipliers, nor any assumption that a sampled Gaussian is feasible off its sample points. The only constraints used for this upper bound follow from the universally quantified hypothesis itself.

The certificate proves $B \leq M$ on the rectangle. The analytic tail bounds are also less than M , and together cover its entire unbounded complement: if $u > 5$ use the first, while if $u \leq 5$ and $w > 7$ then $v > 7$ and the second applies. Therefore $B \leq M < K$ everywhere. This proves Theorem 2, with uniform margin (13).

5 The strict gap and the endpoint condition

For a central-chamber law with $a \geq c$, formula (6) gives $D/2 = ax^2 + c(1 - x^2)$. For every feasible pair (s, ν) in Lemma 3, with $s > 0$, the entropy inequality $H(p) + \sum_i p_i \log h_i \leq \log \sum_i h_i$ gives

$$\begin{aligned}
H(p) + \sum_i p_i \log \nu_i - sD/4 &\leq \log \left(\nu_0 e^{-sx^2/2} + \nu_1 + \nu_2 e^{-s(1-x^2)/2} \right) \\
&\leq \log M < \log K.
\end{aligned} \tag{16}$$

Here $u = \sqrt{s}x$ and $v = \sqrt{s}$ identify the bracket with B after rescaling the reconstruction coordinate. At $s = 0$, feasibility gives $\sum_i \nu_i \leq 1$ and the dual expression is at most zero. Taking the supremum proves $\alpha(P) \leq \log M < \log K$. Reflection handles $c \geq a$. The equalities $a = c$, $a = 1/2$ and $c = 1/2$ are included.

For a self-contained separation from α_3 , take

$$P_* = \frac{157}{200} \delta_0 + \frac{107}{500} \delta_{11/25} + \frac{1}{1000} \delta_1, \quad D(P_*)/4 = \frac{26519}{1250000}, \tag{17}$$

and the dual parameters

$$s = \frac{667}{50}, \quad (\nu_0, \nu_1, \nu_2) = \frac{1}{1000000}(911004, 902898, 985311).$$

The exact endpoint and interpolation calculation below establishes

$$\sup_{y \in \mathbb{R}} \sum_i \nu_i e^{-s(z_i - y)^2} < 0.999988258890 < 1, \quad \sum_i p_i \log(\nu_i/p_i) - sD(P_*)/4 > 0.148821677065. \quad (18)$$

Indeed, for $q = s(z - y)^2 \geq 0$, $|(e^{-s(z-y)^2})''| = 2s|2q - 1|e^{-q} \leq 2s$. For $q \leq 1/2$ this is immediate; for $q \geq 1/2$ use $e^q \geq 1 + q + q^2/2 \geq 2q - 1$. Hence the envelope has second derivative bounded in absolute value by

$$M_2 = 2s \sum_i \nu_i = \frac{1867075071}{25000000}.$$

Evaluate upper enclosures at all 1025 endpoints $j/1024$ and add $M_2/(8 \cdot 1024^2)$ to their maximum, using the linear interpolation remainder. This covers $[0, 1]$. Monotonicity outside this interval covers both infinite tails. Rational logarithm enclosures give the second inequality in (18) and $\log K < 0.14882$. The dual lower bound therefore proves (3). The witness is used only for strict separation; no improvement over the previously reported numerical lower bound is asserted.

We finish by verifying that maximizing laws exist. For $P = (1 - \varepsilon)Q + \varepsilon\delta_z$, where Q is the conditional law on the other two atoms, mixing self-couplings gives $D(P) \geq (1 - \varepsilon)D(Q)$. Reconstruct the exceptional atom exactly and use an almost optimal channel for Q at distortion $D(Q)/4$. Retaining its indicator costs at most $h(\varepsilon)$ in the information upper bound, and discarding it cannot increase information. Thus

$$\alpha(P) \leq h(\varepsilon) + (1 - \varepsilon)\alpha(Q) \leq h(\varepsilon) + (1 - \varepsilon)\alpha_2. \quad (19)$$

In particular, the strict gap (10)–(18) excludes all vanishing-mass boundaries of a maximizing sequence, regardless of relative distance scales.

At a nonconstant limit in finite input coordinates, α is upper semicontinuous. To see this, choose a finite channel within $\eta > 0$ of the limiting infimum, and mix it with exact reconstruction with probability $\rho > 0$. Since $D/4 > 0$, this creates strict distortion slack and increases information by at most $\rho H(P)$. A finite channel can be chosen by quantizing a clipped reconstruction arbitrarily finely before introducing slack. Continue the resulting finite channel at nearby input coordinates, assigning identical rows to labels that coincide at the limit. Distortion, finite-alphabet information and the antitone transport cost are continuous, so the channel remains feasible nearby. Letting η, ρ tend to zero proves upper semicontinuity.

Normalize a maximizing sequence to $(0, x, 1)$ and take a subsequence in the compact coordinate simplex. Equation (19) excludes zero limiting masses. If $x = 0$ or $x = 1$ at the limit, all masses being positive, the limit is a nonconstant binary law; upper semicontinuity and the strict binary gap exclude this possibility. The interior limit is a genuine three-point maximizer. Binary laws are themselves below the strict gap, and (16) excludes all genuine three-point laws with $a, c \leq 1/2$. This completes Theorem 1.

6 Reproduction and limitations

The supplement contains the exact data and verification programs. Run from the **supplement** directory:

```
python3 verify-branch-b.py
python3 certify-gap.py
python3 check-envelope-tails.py
python3 independent-verify.py
```

These programs use only the Python standard library and the supplied data.

For the box bounds, the verifier works at scale $S = 2^{80}$. Given rational $0 \leq r \leq 256$, enclose $r/256$ between adjacent multiples of $1/S$. Recursively enclose its positive Taylor terms, rounding lower products down and upper products up. In the alternating sum, subtract upper terms for the lower sum and lower terms for the upper sum. The degree-18 partial sum bounds $e^{-r/256}$ from above and the degree-19 sum from below, by the signed Taylor remainder. Eight directed interval squarings enclose e^{-r} . Every exponent is range-checked; the maximum possible squared distance in the certified rectangle is 144. No library exponential is used for a proof decision.

For the witness, rational degree-20 and degree-21 sums for $e^{-r/16}$, raised to the sixteenth power, give upper and lower bounds for e^{-r} when $0 \leq r \leq 16$. For logarithms, reduce the positive rational argument to $v \in [1, 2)$ by powers of two and put $t = (v - 1)/(v + 1)$. Then

$$\log v = 2 \sum_{j=0}^{39} \frac{t^{2j+1}}{2j+1} + E, \quad 0 \leq E \leq \frac{2t^{81}}{81(1-t^2)}.$$

Restore the signed multiple of the enclosure for $\log 2$. These formulas certify the tangent, witness and tails; displayed decimals are outward enclosures.

The supplied independent implementation checks all rectangles, coverage, the binary tangent and the witness using a positive exponential series, reciprocal bounds, nine directed squarings at scale 2^{112} , and a different coverage algorithm. Its recorded output is included.

After reflection, every maximizer has $a > 1/2$ and $D/2 = c + bx^2$. The exact maximum and all equality cases remain undetermined here; K is only an exclusion threshold. Primary-source retrieval was unavailable on 8 September 2026. The comparison with [1] is limited to the supplied problem description, and absence of an earlier central-chamber exclusion or endpoint-majority theorem has not been verified.

Computational and AI assistance. AI assisted the analytic arguments, certificate programs and manuscript. Floating-point optimization proposed boxes, multipliers and the witness; all decisive comparisons use exact integers or rationals. A separate model review reconstructed the arguments and replayed the certificates with a separately implemented verifier. This is not human peer review or formal proof-assistant verification.

References

- [1] Arman, Bondarenko, Prymak and Radchenko, *On Grünbaum’s problem for symmetric configurations*, arXiv:2607.22032v1 (2026). <https://arxiv.org/abs/2607.22032v1>.
- [2] Zijian Zeng, Houde Liu, Kuru Ratnavelu, Ong Seng Huat and Yonghua Xiong, *Three Exact Local Positivity Islands and a High-Z Complex-Phase Tube for a Hermitian Rank-Two Scalar Gate*, SSRN (2026). <https://doi.org/10.2139/ssrn.7348478>.